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Econ 297 - Independent Study Proposal

Background

For Spring 2026, I would like to enroll in Independent Study to build upon my prior applied microeconomic research, estimating the causal effect of New York City's 2025 Congestion Pricing Act on motor vehicle collisions within the congestion pricing zone. The original paper uses a difference-in-difference framework, comparing Upper Manhattan (no toll) to Lower Manhattan (toll), validating parallel trends and establishing a weak argument for causality. It was concluded that, on average, the Congestion Pricing Act had a causal effect of 20 fewer accidents per week within the pricing zone. When this paper was written, we only had 10 weeks of post-treatment data to evaluate the average treatment effect, and I was left feeling that the external validity of our estimate was weak.

Project Proposal

I would like to continue to develop this idea through the use of Local Linear Causal Forests (LLCF), developed by Susan Athey and Stephan Wagner at the Stanford Graduate School of Business. The idea is to pivot from an average treatment effect for all of Lower Manhattan, to a high-dimensional map of policy impact across the treatment region. While DiD estimated an average reduction of 20 collisions per week, LLCF would estimate how this reduction varies across different contexts, such as peak versus off-peak hours or specific high-density intersections. With a larger post-treatment sample (62 weeks versus 10 weeks), I hope to identify a robust estimate of the true policy effect since its implementation on January 5th, 2025.

Data

The NYPD publishes weekly batch dumps of accidents that occur within all of New York City. This data includes the date/time/location of each accident, as well as characteristic information of the vehicles involved, such as car type. We also have information on the nature of the crash, such as the reason for the collision, and the number of injuries / deaths reported.

Project Challenges

Optimization Implementation

- LLCF requires solving a Weighted M-Estimation minimization problem for every target prediction point in the map of policy impact. Last quarter I completed *AM 230 - Numerical Optimization* with an A+, where I learned how to formulate these exact types of problems, however, doing so in an applied machine learning context will be new for me.

Standard Errors for Inference

- The original paper uses robust standard errors estimated from the DiD regression. To ensure the p-values are valid under LLCF, I must implement "Honest Trees", where one half of the data determines the splits and the other half estimates the effects. I then must generate standard errors through bootstrap subsampling, which could be computationally expensive when estimating a high-dimensional policy impact map. I may need to use the graduate student social science server, or use parallel processing on my own machine with 64GB of memory.

Feature Engineering for LLCF

- LLCF requires a rich feature set to identify heterogeneous treatment effects. The vehicle collision data from the NYPD may be sufficient, as it contains temporal and spatial data at the scene of each accident. There is a possibility I will need to eventually merge data such as weather at time/location of the accident, subway usage rates in the neighborhood of interest, and holidays that may spike or decrease vehicle usage.